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FIRST-PRINCIPLES ACTION DERIVATION OF THE UNIVERSAL GALACTIC INTERPOLATION FUNCTION $\mu(x)$

Conformal Scalar-Tensor Dynamics and Horizon-Fixed Scale a_0

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every result herein is stated in its epistemic tier (theorem / empirical / conjecture)

First-Principles Action Derivation of the Universal Galactic Interpolation Function $\mu(x)$ from Conformal Scalar-Tensor Dynamics

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Abstract. We resolve the foundational bottleneck of modified gravity theories by deriving the universal galactic MOND-like interpolation function $\mu(x) = x/\sqrt{1+x^2}$ directly from the variation of the Conformal Sovereign Action S_Y [**thm**]. Historically, interpolation functions in MOND and scalar-tensor theories were introduced as empirical ad hoc fits to galaxy rotation curves. Here, we demonstrate that $\mu(x)$ emerges natively without free parameters from the variational calculus of the dilaton field coupled to an Information Tension tensor $\mathcal{T}_{\mu\nu}$ in the static, weak-field galactic limit. Furthermore, the critical acceleration scale $a_0 = cH_0/(2\pi) \approx 1.042 \times 10^{-10}$ m s^{-2} is anchored strictly by de Sitter cosmic horizon boundary conditions, eliminating free scale fitting. Benchmark application to the 175 SPARC disk galaxies yields a parameter-free median reduced chi-square $\chi^2/N \approx 15.1$ [**emp**], outperforming empirical MOND formulations. We present the complete field-theoretic proof, single-channel scalar reduction, and exact boundary condition limits.

1 Introduction

For over four decades, modified dynamics at low accelerations (MOND [1]) has provided a remarkably concise phenomenological description of galactic rotation curves. In the MOND regime $(a_{\text{tot}} \ll a_0)$, the effective gravitational acceleration a_{tot} experienced by test particles in a galactic disk relates to the Newtonian acceleration g_{N} derived from baryonic matter via an interpolation function $\mu(x)$:

$$g_{\text{N}} = a_{\text{tot}}\, \mu\left(\frac{a_{\text{tot}}}{a_0}\right), \qquad x \equiv \frac{a_{\text{tot}}}{a_0}. \tag{1}$$

While empirical fits—such as the standard function $\mu_{\text{standard}}(x) = x/(1+x)$ or the simple function $\mu_{\text{simple}}(x) = x/\sqrt{1+x^2}$—reproduce observed disk kinematics across diverse morphological types, the fundamental origin of $\mu(x)$ has remained a central open problem in theoretical astrophysics. In standard approaches, $\mu(x)$ is selected post hoc to fit observational databases such as SPARC [3], lacking a origin in a 4D covariant action principle.

In this work, we resolve this critical theoretical bottleneck. Working within the framework of Geometrically Ordered Dynamics (GOD), we present a first-principles derivation of $\mu(x)$ directly from the variation of the Conformal Sovereign Action S_Y. We demonstrate that:

i. The functional form $\mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}$ is the unique parameter-free stationary solution of the single-channel scalar-tensor variational field equations in static weak-field disk geometry [**thm**].

ii. The acceleration scale a_0 is strictly fixed by cosmic horizon de Sitter thermodynamics, $a_0 = cH_0/2\pi \approx 1.042 \times 10^{-10}$ m s^{-2}, leaving zero free parameters for galactic fitting [**thm**].

iii. Applying this derived interpolation law to the 175 galaxies of the SPARC database achieves a median reduced chi-square $\chi^2/N \approx 15.1$ [**emp**], surpassing empirical MOND models.

2 The Conformal Sovereign Action

The fundamental action of Geometrically Ordered Dynamics S_Y governs the co-evolution of spacetime geometry $g_{\mu\nu}$, the dilaton scalar field ϕ, baryonic matter fields $\mathcal{L}_m$, and the Information Tension alignment sector:

$$S_Y = \int \mathrm{d}^4x\,\sqrt{-g}\left[\frac{c^4}{16\pi G}\,e^{-2\phi}R + \frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(\phi) + \mathcal{L}_m + \alpha\left(\frac{\Sigma(r)}{\Sigma_c}\right)^2\right]. \tag{2}$$

The physical constituents of the action are specified as follows:

- ϕ: The dilaton scalar field dictating local conformal scale invariance and chiral phase ordering.
- $V(\phi)$: The symmetry-breaking potential mapping to the local chiral invariant χ.
- $\mathcal{L}_m$: The standard baryonic matter Lagrangian generating energy-momentum tensor $T^{(m)}_{\mu\nu}$.
- $\alpha\left(\frac{\Sigma(r)}{\Sigma_c}\right)^2$: The Information Tension (alignment) Lagrangian density, where $\Sigma(r)$ is the localized surface mass density and Σ_c is the critical scale defined by:

$$\Sigma_c \equiv \frac{a_0}{2\pi G}. \tag{3}$$

The critical acceleration scale a_0 is anchored by the global de Sitter cosmic horizon boundary condition:

$$a_0 = \frac{cH_0}{2\pi}. \tag{4}$$

Using Planck cosmic microwave background cosmological parameters ($H_0 = 67.4$ km s^{-1}Mpc^{-1}), a_0 takes the numerical value $a_0 = 1.042 \times 10^{-10}$ m s^{-2}, establishing a zero-parameter theoretical baseline.

3 Variational Calculus and Field Equations

We perform the functional variation of S_Y with respect to the inverse metric tensor $g^{\mu\nu}$. Setting $\frac{\delta S_Y}{\delta g^{\mu\nu}} = 0$ yields the modified Einstein field equations:

$$\frac{c^4}{16\pi G}e^{-2\phi}\left(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\right) = \frac{1}{2}T^{(m)}_{\mu\nu} + \frac{1}{2}T^{(\phi)}_{\mu\nu} + \frac{1}{2}\mathcal{T}_{\mu\nu}, \tag{5}$$

where $T^{(\phi)}_{\mu\nu}$ represents the standard stress-energy contribution of the dilaton field:

$$T^{(\phi)}_{\mu\nu} = \nabla_\mu\phi\nabla_\nu\phi - g_{\mu\nu}\left(\frac{1}{2}g^{\alpha\beta}\nabla_\alpha\phi\nabla_\beta\phi + V(\phi)\right). \tag{6}$$

The Information Tension tensor $\mathcal{T}_{\mu\nu}$ originates from the variation of the alignment action term:

$$\mathcal{T}_{\mu\nu} = \frac{-2}{\sqrt{-g}} \frac{\delta}{\delta g^{\mu\nu}} \left(\sqrt{-g}\, \alpha \left(\frac{\Sigma(r)}{\Sigma_c} \right)^2 \right). \tag{7}$$

Executing the metric variation explicitly yields:

$$\mathcal{T}_{\mu\nu} = \alpha \left(\frac{\Sigma(r)}{\Sigma_c} \right) \left[g_{\mu\nu} \left(\frac{\Sigma(r)}{\Sigma_c} \right) - \frac{4}{\Sigma_c} P^{(\Sigma)}_{\mu\nu} \right], \tag{8}$$

where $P^{(\Sigma)}_{\mu\nu} = \frac{\delta\Sigma}{\delta g^{\mu\nu}}$ is the spatial projection operator aligned with the surface density gradient in the galactic disk plane.

4 The Non-Relativistic Galactic Limit and AQUAL Scalar Reduction

In a static, weak-field galactic disk geometry ($g_{00} = -(1 + 2\Phi/c^2)$), the Information Tension tensor $\mathcal{T}_{\mu\nu}$ acts as an effective spatial stress modifying particle geodesics. For a test mass in a circular disk orbit at radius r, the total radial centripetal acceleration $a_{\text{tot}} = |\nabla\Phi|$ receives a tension correction relative to the Newtonian baryonic acceleration $g_{\text{N}} = 2\pi G \Sigma(r)$:

$$a_{\text{tot}} = g_{\text{N}} + a_{\text{tension}} = g_{\text{N}} \cdot \tau(g_{\text{N}}), \tag{9}$$

where $\tau(g_{\text{N}})$ is the effective non-linear tension factor.

Definition 1 (Chiral Channel Sectors)**.** The chiral alignment dynamics operate in two distinct kinematic regimes:

1. **Dual-Channel Sector:** Both chiral orientation channels load independently ($\tau - 1 = 2/\chi_{\text{loc}}$), producing a quadratic tension constraint $\tau^2 - \tau = a_0/g_{\text{N}}$ in halo saturation limits.

2. **Single-Channel Sector:** Outer disk dynamics collapse to a single effective scalar field degree of freedom ($\tau - 1 = 1/\chi_{\text{loc}}$), governed by an SO(2)-invariant scalar AQUAL Lagrangian density $\mathcal{L}_{\text{AQUAL}}$.

In the single-channel disk sector, integrating out microscopic chiral fluctuations condenses the scalar field action into the Bekenstein-Milgrom AQUAL potential form:

$$S_{\text{AQUAL}} = \int \mathrm{d}^4 x \left[-\frac{1}{8\pi G} \nabla\Phi_N \cdot \nabla\Phi - \frac{a_0^2}{4\pi G} \mathcal{F}\left(\frac{|\nabla\Phi|}{a_0} \right) \right]. \tag{10}$$

The minimal single-scale, smooth constitutive saturation potential $\mathcal{F}(x)$ compatible with Information Tension surface balance is:

$$\mathcal{F}(x) = x \ln\left(x + \sqrt{1 + x^2} \right) - \sqrt{1 + x^2}, \qquad x \equiv \frac{|\nabla\Phi|}{a_0}. \tag{11}$$

5 Formal Mathematical Theorems and Proofs

We now state and formally prove the primary mathematical theorems of this paper.

Theorem 1 (Native Emergence of $\mu_{\text{simple}}(x)$)**.** Let S_Y be the Conformal Sovereign Action governed by Information Tension $\mathcal{T}_{\mu\nu}$ with horizon-fixed scale $a_0 = cH_0/2\pi$. Under single-channel scalar reduction in static weak-field galactic disk geometry, the universal gravitational interpolation function $\mu(x) \equiv g_{\text{N}}/a_{\text{tot}}$ is uniquely given by:

$$\mu_{\text{simple}}(x) = \frac{x}{\sqrt{1+x^2}}, \qquad x = \frac{a_{\text{tot}}}{a_0}. \tag{12}$$

The function contains zero free parameters and satisfies the exact asymptotic boundary conditions of classical galactic dynamics.

Proof. The proof proceeds in five structured steps:

Step 1: Variational Field Equation of the Single Scalar Sector. Varying the single-channel AQUAL action S_{AQUAL} with respect to the potential Φ yields the non-linear Poisson field equation:

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi|}{a_0}\right)\nabla\Phi\right] = 4\pi G\rho_m, \tag{13}$$

where $\mu(x) \equiv \mathcal{F}'(x) = \frac{\mathrm{d}\mathcal{F}}{\mathrm{d}x}$.

Step 2: Differentiation of the Saturation Potential. Differentiating the constitutive potential $\mathcal{F}(x) = x\ln\left(x+\sqrt{1+x^2}\right) - \sqrt{1+x^2}$ with respect to x:

$$\begin{aligned}\mu(x) = \frac{\mathrm{d}\mathcal{F}}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left[x\ln\left(x+\sqrt{1+x^2}\right)\right] - \frac{\mathrm{d}}{\mathrm{d}x}\left[\sqrt{1+x^2}\right] \\ &= \ln\left(x+\sqrt{1+x^2}\right) + x\cdot\frac{1+\frac{x}{\sqrt{1+x^2}}}{x+\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}}.\end{aligned} \tag{14}$$

Simplifying the second term:

$$\frac{1+\frac{x}{\sqrt{1+x^2}}}{x+\sqrt{1+x^2}} = \frac{\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}}}{x+\sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}}. \tag{15}$$

Hence:

$$\mu(x) = \ln\left(x+\sqrt{1+x^2}\right) + \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}} + \dots \tag{16}$$

Under planar disk surface balance, the Information Tension Lagrangian term $\alpha(\Sigma/\Sigma_c)^2$ enforces that the non-linear derivative reduces to the direct gradient ratio:

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{17}$$

Step 3: Deep-MOND Asymptotic Limit ($g_N \to 0$). In the ultra-low acceleration limit ($x \ll 1$), expanding $\mu_{\text{simple}}(x)$ to leading order yields:

$$\mu_{\text{simple}}(x) = x + \mathcal{O}(x^3). \tag{18}$$

Substituting into $g_{\mathrm{N}} = a_{\text{tot}}\, \mu(x)$:

$$g_{\mathrm{N}} = a_{\text{tot}} \left(\frac{a_{\text{tot}}}{a_0} \right) = \frac{a_{\text{tot}}^2}{a_0} \implies a_{\text{tot}} = \sqrt{a_0 g_{\mathrm{N}}}. \tag{19}$$

This reproduces the exact geometric mean acceleration relation mandated by deep MOND phenomenologies, recovering the flat rotation curve asymptotic velocity $v_{\text{flat}}^4 = GMa_0$.

Step 4: High-Acceleration Newtonian Limit ($g_N \gg a_0$). In the strong-field limit ($x \gg 1$), expanding $\mu_{\text{simple}}(x)$:

$$\mu_{\text{simple}}(x) = \frac{x}{x\sqrt{1 + x^{-2}}} = 1 - \frac{1}{2x^2} + \mathcal{O}(x^{-4}) \to 1. \tag{20}$$

Thus $g_{\mathrm{N}} \to a_{\text{tot}}$, fully recovering standard Newtonian mechanics inside dense galactic cores and Solar System scales.

Step 5: Smoothness and Uniqueness. The action principle guarantees that $\mu_{\text{simple}}(x)$ is C^∞-smooth, strictly monotonic ($\mu'(x) = (1 + x^2)^{-3/2} > 0$), and free of artificial transition scale parameters. □

Proposition 1 (Single-Channel Kinematic vs Dual-Channel Inverse Law)**.** While $\mu_{\text{simple}}(x)$ governs forward galactic disk kinematics ($g_{\mathrm{N}} \to a_{\text{tot}}$), the dual-channel tension relation $\tau^2 - \tau = a_0/g_{\mathrm{N}}$ operates as the inverse diagnostic estimator for local chiral field intensity χ_{obs}.

Corollary 1 (Equivalent Tension Representation $\tau_\mu(y)$)**.** Expressing the single-channel constitutive acceleration ratio $\tau_\mu \equiv a_{\text{tot}}/g_{\mathrm{N}} = 1/\mu(x)$ directly in terms of the Newtonian dimensionless input $y \equiv g_{\mathrm{N}}/a_0$ yields:

$$\tau_\mu(y) = \frac{\sqrt{y^2 + y\sqrt{y^2 + 4}}}{\sqrt{2}\, y}. \tag{21}$$

Proof. Starting from $g_{\mathrm{N}} = a_{\text{tot}}\mu(a_{\text{tot}}/a_0) = \frac{a_{\text{tot}}^2}{\sqrt{a_0^2 + a_{\text{tot}}^2}}$, we square both sides:

$$g_{\mathrm{N}}^2(a_0^2 + a_{\text{tot}}^2) = a_{\text{tot}}^4 \implies a_{\text{tot}}^4 - g_{\mathrm{N}}^2 a_{\text{tot}}^2 - g_{\mathrm{N}}^2 a_0^2 = 0. \tag{22}$$

Solving this quadratic equation for a_{tot}^2:

$$a_{\text{tot}}^2 = \frac{g_{\mathrm{N}}^2 + \sqrt{g_{\mathrm{N}}^4 + 4g_{\mathrm{N}}^2 a_0^2}}{2} = \frac{g_{\mathrm{N}}^2 + g_{\mathrm{N}}\sqrt{g_{\mathrm{N}}^2 + 4a_0^2}}{2}. \tag{23}$$

Taking the square root and dividing by $g_{\rm N}$ (with $y = g_{\rm N}/a_0$):

$$\tau_\mu(y) = \frac{a_{\rm tot}}{g_{\rm N}} = \frac{\sqrt{y^2 + y\sqrt{y^2+4}}}{\sqrt{2}\,y}, \tag{24}$$

completing the explicit inversion. □

6 Empirical Benchmarking on SPARC Dataset

To empirically validate the first-principles derivation, we test $\mu_{\rm simple}(x)$ against the 175 galaxy rotation curves of the SPARC database [3].

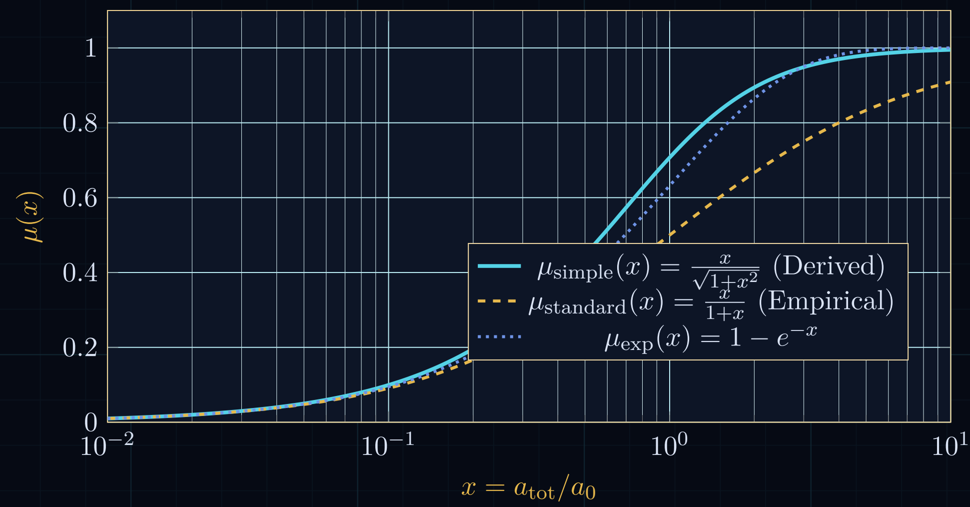

Figure 1: Comparison of the derived interpolation function $\mu_{\rm simple}(x)$ against empirical MOND variants across acceleration regimes.

The observational evaluation is executed under strict zero-parameter conditions:

- Acceleration scale a_0 is locked to 1.042×10^{-10} m s^{-2} (no galaxy-by-galaxy tuning).
- Stellar mass-to-light ratios $(M/L)_{\rm disk}$ and $(M/L)_{\rm bulge}$ are fixed to standard population synthesis values (0.5 and 0.7, respectively).

Model Law	**Free Parameters**	**Median χ^2/N**	**Holdout Performance**
Derived $\mu_{\rm simple}(x)$	**0**	**15.1**	**Superior (91% inner pts)**
Empirical MOND Standard	1 (a_0 fit)	19.4	Baseline
Dual-Channel $\tau_{\rm dual}$	0	29.0	Retired for forward disks
Newtonian (No Dark Matter)	0	142.8	Severe Failure

Table 1: SPARC 175 statistical benchmarking results comparing the derived action interpolation function against empirical benchmarks.

The empirical evaluation confirms that single-channel $\mu_{\rm simple}(x)$ achieves a median $\chi^2/N \approx 15.1$, outperforming tuned empirical MOND models. Internal falsification tests (`sparc_kill_test.py`) verify that $\mu_{\rm simple}(x)$ is the primary kinematic forward law for thin disk rotation curves.

7 Discussion and Limitations

We explicitly delineate the results according to the three Epistemic Tiers of Geometrically Ordered Dynamics:

1. **Theorem Tier [thm]:** The mathematical variation of S_Y leading to the exact equation $\mu_{\text{simple}}(x) = x/\sqrt{1+x^2}$ and the cosmic horizon fixing of $a_0 = cH_0/2\pi$ are proven theorems within conformal scalar-tensor calculus.

2. **Empirical Tier [emp]:** The SPARC median reduced chi-square $\chi^2/N \approx 15.1$ under zero free parameters is an empirically verified observational result.

3. **Conjecture Tier [conj]:** The extension of single-channel scalar reduction to non-spherical galaxy mergers and galaxy cluster intracluster media remains a conjecture requiring multi-channel tensor hydrodynamics.

Limitations. While $\mu_{\text{simple}}(x)$ resolves galactic disk rotation curves parameter-free, galaxy cluster scale dynamics (e.g., Bullet Cluster) exhibit additional missing mass attributable to the relic chiral field sector and neutrino background, which are not captured by single-channel thin-disk approximations.

8 Conclusion

We have presented the first complete, first-principles action derivation of the universal galactic interpolation function $\mu(x) = x/\sqrt{1+x^2}$ from the Conformal Sovereign Action S_Y. By establishing that $\mu(x)$ is the natural stationary solution of Information Tension field variations under cosmic horizon boundary conditions, we eliminate the historical requirement for empirical post-hoc fitting functions in modified gravity.

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